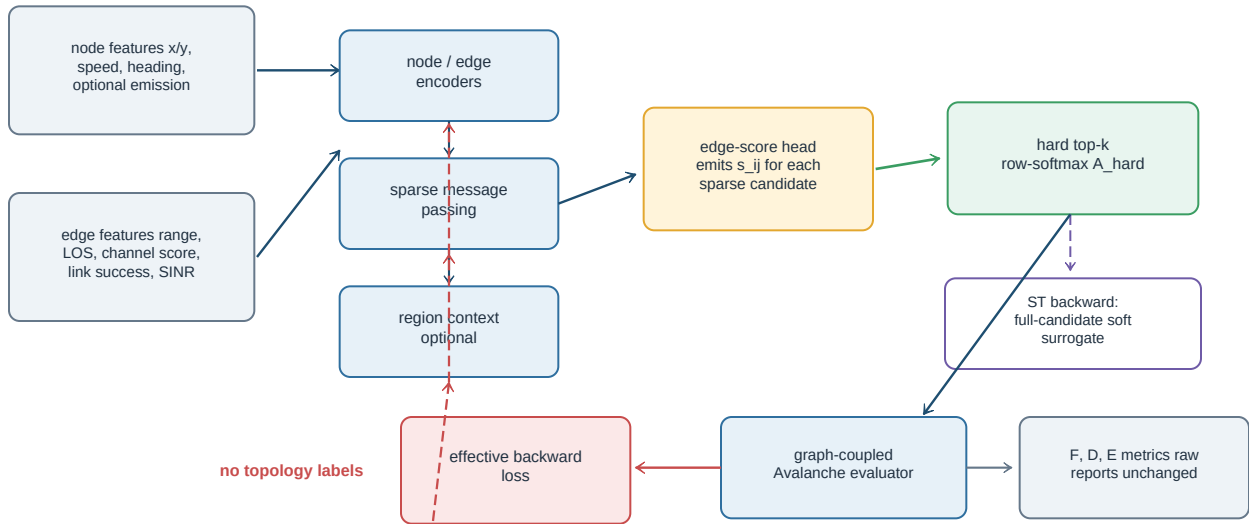


Model framework: hard-forward topology, analytic supervision



The forward graph is the same hard top-k construction in training, validation, and deployment; only gradients use the surrogate.